

Introduction to Knot Theory

(4) Seifert Surface and Alexander Module

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The Infinite Cyclic Cover

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$$\tilde{X} \longrightarrow X$$

the infinite cyclic cover, with deck transformation $t : \tilde{X} \rightarrow \tilde{X}$.

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Alexander Module

$$A_K := H_1(\tilde{X}; \mathbb{Z})$$

as a $\mathbb{Z}[t, t^{-1}]$ -module, where t acts by the deck transformation.

Chain Complex of the Cyclic Cover

The CW-structure of X lifts to $\tilde{X}$:

$$\cdots \longrightarrow C_2(\tilde{X}) \xrightarrow{\partial_2} C_1(\tilde{X}) \xrightarrow{\partial_1} C_0(\tilde{X}) \longrightarrow 0$$

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Key Point

$$A_K = \ker(\partial_1) / \operatorname{im}(\partial_2) = \operatorname{coker}(\partial_2 : C_2 \rightarrow \ker \partial_1)$$

Everything is determined by ∂_2 .

Fox Calculus

From a presentation of $\pi_1(X)$ with generators $x_1, \dots, x_\mu$ and relators $r_1, \dots, r_\nu$:

$$(\partial_2)_{ij} = \phi\left(\frac{\partial r_i}{\partial x_j}\right) \in \mathbb{Z}[t, t^{-1}]$$

where $\partial/\partial x_j$ is the Fox free derivative and $\phi : \mathbb{Z}[\pi_1] \rightarrow \mathbb{Z}[t, t^{-1}]$ is the abelianization.

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Goal

∂_2 is defined algebraically via Fox calculus.

We seek a geometric construction that produces the same map.

Seifert Surface

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- The orientation of F induces an orientation on ∂F , which must agree with that of K .
- F is embedded (no self-intersections) in S^3 .

Existence Theorem

Theorem (Seifert, 1934)

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Constructed explicitly via the **Seifert algorithm** from any diagram of K . □

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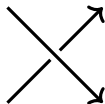
- Existence is not obvious: non-orientable spanning surfaces always exist, but we need orientability.
- For a link, the surface can always be chosen to be connected.

Seifert Algorithm: Step 1 – Smoothing

Step 1: Smooth every crossing

At each crossing, replace it with the **oriented smoothing**.

positive crossing



oriented smoothing

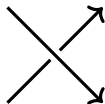


Seifert Algorithm: Step 1 – Smoothing

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- Replace each crossing with two arcs respecting the orientation.
- Do this for every crossing simultaneously.

Seifert Algorithm: Steps 2–3

Step 2: Seifert circles

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- Fill each Seifert circle with a disk.
- At each original crossing, attach a **half-twisted band** connecting the two adjacent disks.
- The twist direction is determined by the sign of the crossing.

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The result is a compact orientable surface F with $\partial F = K$.

Example of trefoil knot (3_1)

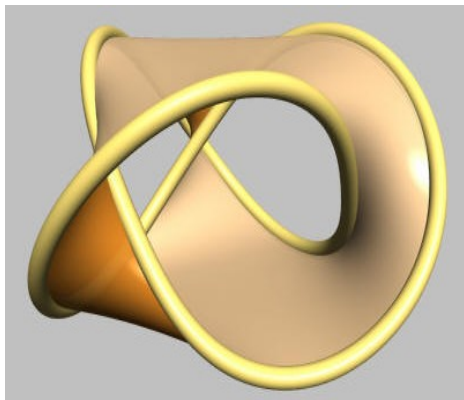


Figure: Seifert surface of trefoil knot

https://vanwijk.win.tue.nl/seifertview/knot_gallery.htm

Euler Characteristic and Genus

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F has s disks (contributing $+s$) and c bands (contributing $-c$).

For a connected orientable surface with one boundary component: $\chi(F) = 1 - 2g(F)$. Hence:

$$g(F) = \frac{c - s + 1}{2}$$

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- The algorithm does not always produce a minimal-genus surface.
- Genus is a knot invariant, but proving minimality requires more work.

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Example

For the trefoil 3_1 : $s = 2$ circles, $c = 3$ crossings, so $g(F) = (3 - 2 + 1)/2 = 1$.

Homology of the Seifert Surface

Fix a Seifert surface F of genus g with one boundary component:

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Push-off

Since $F \subset S^3$, the surface has a well-defined normal direction. For a curve a on F , define the **positive push-off** a^+ by pushing a slightly in the positive normal direction.

Seifert Pairing

Definition

The **Seifert pairing** is the bilinear form

$$\theta : H_1(F) \times H_1(F) \longrightarrow \mathbb{Z}, \quad \theta(a, b) = \text{lk}(a^+, b),$$

where lk denotes the linking number in S^3 .

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where lk denotes the linking number in S^3 .

- Well-defined: a^+ is disjoint from F , hence from $b \subset F$.
- **Not symmetric** in general: $\text{lk}(a^+, b) \neq \text{lk}(a, b^+)$.
- Encodes how curves on F link with push-offs of other curves in S^3 .

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For any curves a, b on F :

$$\text{lk}(a^+, b) - \text{lk}(a^-, b) = (a \cdot b)_F,$$

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Note

V depends on the basis and Seifert surface, but its *S-equivalence class* is a knot invariant.

From Seifert Matrix to the Boundary Map

Theorem

Under the identification of $H_1(F)$ with a subspace of $H_1(\tilde{X})$:

$$\partial_2 = tV - V^T$$

as a $\mathbb{Z}[t, t^{-1}]$ -module map $\mathbb{Z}[t, t^{-1}]^{2g} \rightarrow \mathbb{Z}[t, t^{-1}]^{2g}$.

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Idea:

- $\tilde{X} \sim \dots \cup t^{-1}F \cup F \cup tF \cup \dots$, so $H_1(\tilde{X}) \cong H_1(F) \otimes \mathbb{Z}[t, t^{-1}]$.
- Essential informations are all in $2g$ generators; eliminate linearly dependent terms of generator in Wirtinger presentation.
- $\text{lk}(a_i^+, a_j)$ are each on adjacent sheet (upper(t) sheet); $\text{lk}(a_i^-, a_j)$ are in same sheet.

Alexander Module and Polynomial

Corollary

$$A_K = \operatorname{coker}(\partial_2) = \operatorname{coker}(tV - V^T) = \frac{\mathbb{Z}[t, t^{-1}]^{2g}}{(tV - V^T) \mathbb{Z}[t, t^{-1}]^{2g}}$$

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$$\Delta_K(t) = \det(tV - V^T) \in \mathbb{Z}[t, t^{-1}],$$

well-defined up to units $\pm t^k$.

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Alexander Polynomial

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well-defined up to units $\pm t^k$.

- The polynomial arises from a **bilinear (Seifert) structure**.
- Symmetry: $\Delta_K(t) \doteq \Delta_K(t^{-1})$ follows from $\det(tV - V^T) = (-t)^{2g} \det(t^{-1}V^T - V) = t^{2g} \det(t^{-1}V - V^T)$.

Dependence on Choices

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Change of Surface

Different Seifert surfaces can give non-congruent matrices. We need a coarser equivalence relation.

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A stabilization replaces F by $F' = F \natural_{\partial} T$ (boundary connected sum with a handle), enlarging the matrix:

$$V \longmapsto \begin{pmatrix} V & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & 0 & 0 \\ \mathbf{0} & 1 & 0 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} V & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & 0 & 1 \\ \mathbf{0} & 0 & 0 \end{pmatrix}.$$

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S-equivalence

Two Seifert matrices are **S-equivalent** if they are related by:

- Congruence: $V \sim PVP^T$ for $P \in GL(\mathbb{Z})$,
- Stabilization and its inverse (destabilization).

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- In particular, $\det(tV - V^T)$ is well-defined up to $\pm t^k$.
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Remark

S-equivalence is *strictly coarser* than congruence. Two distinct knots can have S-equivalent Seifert matrices.

The Alexander polynomial is one kind invariant from S-equivalence class of seifert matrix.

Knot Signature

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Example

For the trefoil 3_1 : $V = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}$, so $V + V^T = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}$ has eigenvalues $-1, -3$, giving $\sigma(3_1) = -2$.

References

- D. Rolfsen, *Knots and Links*, AMS Chelsea.
- A. Hatcher, *Algebraic Topology*, Cambridge Univ. Press.

Images

- Rolfsen, *Knots and Links*
- Jack van Wijk personal site (Seifert Surface)